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|------------------------------------------------|--------------------------------------|
| Program : <b>Diploma in Polymer Technology</b> |                                      |
| Course Code : <b>4093</b>                      | Course Title: <b>Rubber Products</b> |
| Semester : <b>4</b>                            | Credits: <b>4</b>                    |
| Course Category: <b>Program Core</b>           |                                      |
| Periods per week: <b>4 (L:3, T:1, P:0)</b>     | Periods per semester: <b>60</b>      |

### Course Objectives:

- To acquire the knowledge of different rubber products in the industry.
- To write recipe for these products.
- To understand the manufacture of moulded, extruded, calendered, miscellaneous rubber products To impart knowledge on rubber to metal bonded products, rubber coated and assembled rubber products.

### Course Prerequisites:

| Topic                                | Course code | Course name               | Semester |
|--------------------------------------|-------------|---------------------------|----------|
| Marketable forms of Natural Rubber   |             | Natural rubber production | 3        |
| Synthetic rubber, properties         |             | Synthetic rubber          | 3        |
| Rubber compounding and vulcanisation |             | Technology of elastomers  | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn | Description                                                                       | Duration (Hours) | Cognitive level |
|-----|-----------------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Paraphrase moulded and expanded rubber products                                   | 15               | Understanding   |
| CO2 | Generalize extruded, calendered and miscellaneous rubber products                 | 14               | Understanding   |
| CO3 | Explain rubber to metal bonded rubber products and rubber coated textile products | 15               | Understanding   |
| CO4 | Describe assembled rubber products                                                | 14               | Understanding   |
|     | Series Test                                                                       | 2                |                 |

**CO - PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     | 3   |     |     |     |     |
| CO2             | 3   |     | 3   |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description                                                                             | Duration (Hours) | Cognitive level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Paraphrase moulded and expanded rubber products</b>                                  |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comprehend the production of compression moulded products and its finishing operations. | 3                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outline the chemistry and production of hard rubber.                                    | 4                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Explain the classification and production of expanded rubber products                   | 4                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Identify the major defects and remedies of compression moulded products                 | 4                | Understanding   |
| <b>Contents:</b><br>Steps involved in production of compression moulded products - Automobile bushes - Door mats, IB Caps, Hot water bags - Chemistry of hard rubber products (Ebonite). Method of production for battery boxes - Finishing of rubber mouldings. Expanded Products: Types of blowing agents and their comparison - Compounding and Molding of Cellular rubber products, MC, Hawaii, Sponge - Typical formulations. Defects and remedies of molded products. |                                                                                         |                  |                 |
| <b>CO2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Generalize extruded, calendered and miscellaneous rubber products</b>                |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Describe the manufacture of extruded products.                                          | 4                | Understanding   |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Summarize the manufacture of calendered products                                        | 4                | Understanding   |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Illustrate the steps involved in the spreading process                                  | 2                | Understanding   |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Explain the adhesive treatment for nylon , nylon6 and polyester                         | 4                | Understanding   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Series test I                                                                           | 1                |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                    |   |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>Contents</b><br><b>Extruded rubber products</b><br>Automobile treads, - channels -, inner tubes, Garden hose - Typical formulations - Methods of manufacture.<br><b>Calendered rubber products</b><br>Supported and unsupported sheeting- Hospital sheeting, -production method -Profile calendered products. Production of supported sheets. Defects and remedies of the above.<br>Rubber coated textile products<br>Tarpaulin and other rubber coated textile, rubber lining, spreading.<br>Design principles of spreading, compounding and typical formulations.<br>Adhesive treatment for Nylon 6 - Nylon 66 - Polyester - Adhesive composition |                                                                                                                                    |   |               |
| :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                    |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Explain rubber to metal bonded products and miscellaneous rubber products</b>                                                   |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Outline the principle and steps involved in the production of rubber to metal bonded products .                                    | 3 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Explain the manufacture of play ball, tennis ball and golf ball with special reference to formulations and principle of expansion. | 4 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Discuss the manufacture of gaskets and seals                                                                                       | 4 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Describe the manufacture of different types of rubber footwears.                                                                   | 4 | Understanding |
| <b>Contents:</b><br>Rubber to Metal bonded products - Principle of bonding rubber to metal - bonding methods - List major rubber to metal bonded products - rubber covered rollers - type writer roller - paper mill roller - textile mill roller - rice polishing roller - compound preparation, assembling curing, finishing Sports Goods: Play ball - Tennis balls - Components - method of manufacture, typical formulations. Oil seals, Gaskets-Manufacture and Typical formulations. Footwear - Different methods fo manufacture- compression moulded - injection moulded - assembled.                                                           |                                                                                                                                    |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Describe assembled rubber products</b>                                                                                          |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Explain the manufacture of Conveyor and Transmission and V- belts.                                                                 | 5 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Explain the manufacture of different types of hoses.                                                                               | 5 | Understanding |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---|---------------|
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Describe the manufacture of different types of cables | 4 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Series test II                                        | 1 |               |
| <b>Contents:</b><br>Assembled rubber products. Beltings: Classification - Conveyor belts - Transmission and V - belts: components manufacturing techniques, Batch and continuous curing, Compounding, Typical formulations. Hoses: Classification according to manufacture and end use. Components. Molded hose, Mandrel cured hose - comparison. Braided hose - Handmade hose, Radiator hose, Petrol hose, Sand blasting hose, LPG tube - Typical formulations -method of manufacture. Cables: Classification, Power cables and their components, manufacturing methods, Curing- Typical formulations. |                                                       |   |               |

#### **Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                        |
|------------|-------------------------------------------------------------------------------------------------|
| T1         | C.M.Blow. Rubber Technology And Manufacture Publisher - Butterworth Hainemann Ltd,1982 Edition. |
| R1         | Maurice Morton. Rubber Technology, Publisher - Springer, 1999 Edition.                          |
| R2         | R.T Vanderbilt .Vanderbilt Rubber Hand book, Publisher - R.T Vanderbilt, 1990 Edition.          |
| R3         | IRI, Rubber engineering - McGraw - Hill.                                                        |
| R4         | Anil K Bhowmick, Rubber Product Manufacturing Technology -                                      |

#### **Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                     |
|--------------|-------------------------------------------------------------------------|
| 1            | <a href="https://www.jacionline.org">https://www.jacionline.org</a>     |
| 2            | <a href="http://www.nocil.comLatex">http://www.nocil.comLatex</a>       |
| 3            | <a href="http://www.rubberscience.in">http://www.rubberscience.in</a> . |
| 4            | <a href="https://www.researchgate.net">https://www.researchgate.net</a> |
| 5            | <a href="http://www.iri.net.in">www.iri.net.in</a>                      |